

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-3 Semester (2025-26)
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Branch	CSE
Course Outcome	4

Case Study Topic: Hyderabad Metro Route Optimization using Dijkstra's Algorithm with Line Transfer Penalties

Word Done Summary:

In this case study, Dijkstra's Algorithm was applied to determine the shortest travel time in the Hyderabad Metro network. Unlike traditional shortest-path problems, metro transportation involves **line-transfer penalties** whenever passengers switch between metro lines. The metro network consists of seven stations:

LBN = LB Nagar
AMP = Ameerpet
BGP = Begumpet
KKP = Kukatpally
HTC = Hitec City
MHM = MehdiPatnam
SCB = Secunderabad

Three metro lines are available:

Red Line

LBN → AMP → MHM → SCB

Blue Line

AMP → BGP → KKP → HTC

Green Line

AMP → KKP

Transfer penalty:

5 minutes

Implementation Details:

1. Without transfer penalty.
2. With transfer penalty.
3. Using graph transformation with virtual nodes.
4. Analyze real-world metro route planning.

Result:

Without Transfer Penalty

Path: LBN → AMP → KKP → HTC

Time: 35 Minutes

With Transfer Penalty

Path: LBN → AMP → BGP → KKP → HTC

Time: 41 Minutes

Dijkstra's Algorithm successfully computed the shortest travel route in the Hyderabad Metro network. By transforming interchange stations into virtual line-specific nodes and introducing transfer edges, transfer penalties were naturally incorporated into the shortest-path computation. This approach is widely used in real-world transit systems because it separates route computation from transfer logic and allows standard shortest-path algorithms to be applied efficiently.

Output:

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Shortest Time = 35 minutes
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GitHub Link:

<https://github.com/saanvikompelli-cmd/DSA-2-Case-Studies>

Student Signature	Faculty Signature
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